Mobile Applications and Technologies (2026/2027)

Laboratory Project

Monitoring and Alarm System

(Part 2 – Contiki)

1 Introduction

Mobile technologies and applications using them is an area with increasing importance every day. Many application fields can benefit from this situation. There are many different designations in the literature related to this subject (e.g. Internet of Things (IoT); Cyber-physical systems; Wireless Sensor Networks (WSN)), corresponding to situations of complementarity or overlapping, but, in a generic way, and abstracting some particularities, we can look at this scenario as one where we have "smart objects" that can be interconnected. In some cases making WSNs, and in other cases having a more direct use or interaction.

This part of the laboratory project has as main goal the familiarization of students with the development of applications that use Wireless Sensor Networks.

2 Problem description

This part of the project corresponds to a WSN application ("Monitoring and Alarm System") that runs on "motes" with the support of the Contiki Operating System. This type of application can do the monitoring and control of several parameters (e.g. temperature, light, people presence) associated with rooms/sectors of a building/factory. In this particular case, we want to monitor the temperature and people presence in several rooms of a building. More precisely the application should be able to:

- For each room, at local (node) level:
  - monitor the temperature, and indicate if it is above a specified threshold (use red LED);
  - detect people presence, and depending on that, turn on or off lights (use yellow LED);
  - signal activity by blinking (toggling every second) an LED (use green LED);
- if enabled, report to the sink node the high temperature situation and/or people detection (alarm notification);
- keep at the sink node the last 5 notifications, and allow its consultation;
- allow the modification of temperature threshold and the alarm notification enabling/disabling;
- [interact with project part1 or network server] (**open possibility**).

3 Project development

The application is programmed using the C programming language and will have the support of the Contiki Operating System. It will run using the Cooja Simulator. Students may choose the type of mote to use (from the ones available in the Cooja environment).

It is students responsibility to structure the program in a modular fashion according to the several tasks that must be performed on the several nodes. Most of the project organization is a project decision not *a priori* imposed, but it should take into account the main mechanisms offered by the Contiki ecosystem. It should deal with: sensor integration (even if they are "emulated" in the case of Cooja environment); collection and processing of data, and sending of relevant information to the sink node; actuation (e.g. alarm generation, turn on/off lights (simulated using the LEDs)); user interaction at sink node ("console" using serial port). The application should have more than 1 node, and at least 1 node with sensors/actuators and several activities.

In the development of the project, the utilization of a modular structure and a phased testing is advised.

4 Project Reports

Project Reports should be structured in such a way that the main subjects addressed in the course should be highlighted (as much as possible):

- Application description (characteristics / requirements);
- Node Architecture and Network Architecture (both hardware and software);
- [Operating] System Support and Special Requirements;
- Network Communications;
- Sensors and Actuators;
- Energy Management

5 Project delivery

This part of the project should be delivered by **18/12/2026**. The delivery consists of a digital copy (ZIP file) containing all developed programs and a small report (3 or 4 pages) explaining the main components of the application, and their interactions. All these elements should be correctly identified (group number and students). Project demonstrations, based on the delivered material, and oral exams, related to both parts of the project, will be done in the week starting on 04/01/2027.